

ReviewAgent AI

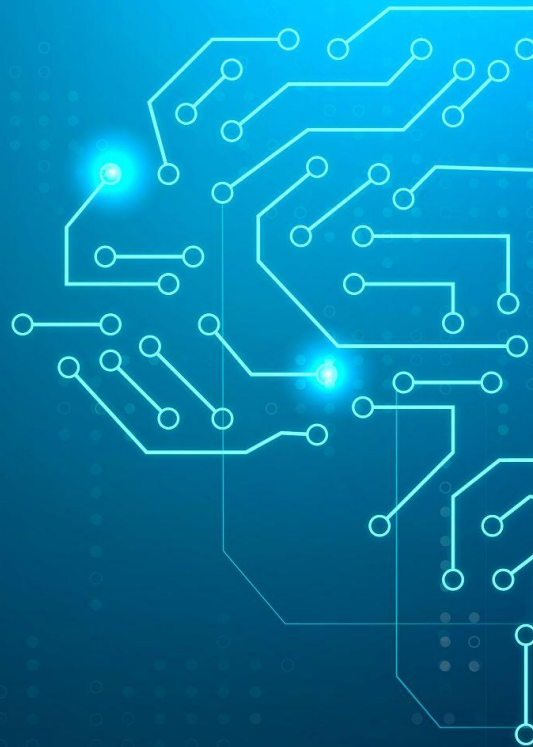
AI-Powered Pull Request Security Auditor

Reasoning-based security reviews for modern development teams —
catching vulnerabilities before they reach production.

DEVELOPED BY PRAVEEN G

AZURE OPENAI

GITHUB INTEGRATION



The Security Review Bottleneck

Manual code reviews cannot keep pace with modern development velocity — leaving critical vulnerabilities undetected until it is too late.



Slow Reviews

Manual PR reviews create deployment delays and developer frustration.



Missed Vulnerabilities

Human reviewers overlook subtle security flaws under time pressure.



Cannot Scale

Security teams are overwhelmed as code volume grows exponentially.



Inconsistent Quality

Review standards vary by reviewer, team, and time of day.



Production Risk

Undetected flaws reach production, exposing the business to breaches.

Introducing ReviewAgent AI

An intelligent, reasoning-based security layer that integrates directly into your pull request workflow — delivering instant, consistent, and actionable security feedback.



Every pull request is automatically analyzed, scored, and enriched with fix recommendations — no manual triage required.



Why ReviewAgent AI Matters

70%

Faster Reviews

Security analysis delivered in seconds, not hours.

90%

Faster Discovery

Vulnerabilities identified earlier in the development cycle.

85%

Security Coverage

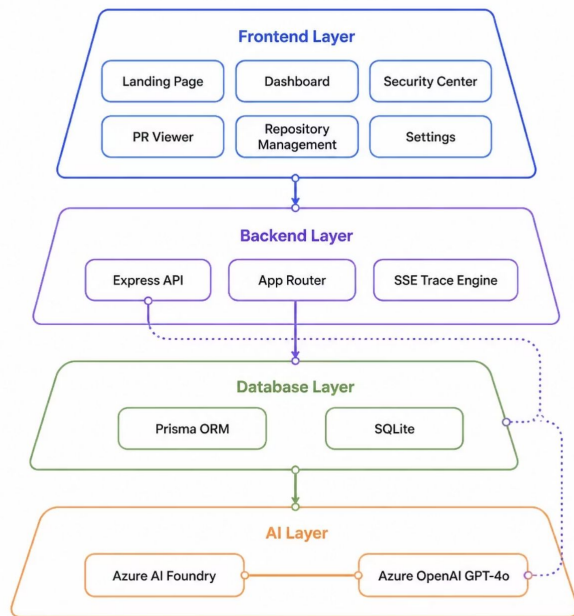
Comprehensive analysis across OWASP Top 10 and beyond.

75%

Less Manual Effort

Security teams focus on high-risk issues, not routine scans.

System Architecture



Four-Layer Design

ReviewAgent AI is built on a clean, modular architecture that separates concerns across four distinct layers — ensuring maintainability, scalability, and seamless AI integration.

Frontend

React-based UI with real-time dashboards

Backend

Express API with SSE trace engine

Database

Prisma ORM + SQLite for data persistence

AI Engine

Azure AI Foundry + GPT-4o for reasoning

End-to-End Application Workflow

From pull request to security report — ReviewAgent AI automates the entire review pipeline with zero friction for developers.

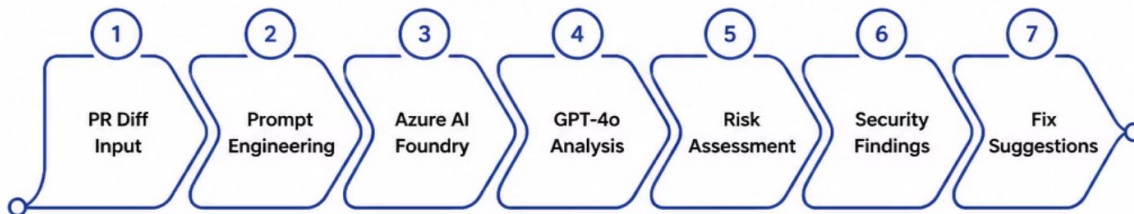


Each stage is powered by Azure OpenAI GPT-4o, ensuring deep reasoning and context-aware security analysis at every step.

Azure AI Security

Powered by Azure OpenAI GPT-4o

ReviewAgent AI leverages Azure AI Foundry and GPT-4o to perform deep, reasoning-based security analysis — not just pattern matching. The engine understands code context, intent, and risk surface.



→ Prompt Engineering

Custom-crafted prompts for precise vulnerability detection

→ Risk Assessment

Multi-dimensional scoring for severity and confidence

→ Fix Suggestions

Actionable, code-level remediation guidance

Dual AI Execution Architecture

ReviewAgent AI supports both demo and live modes with automatic switching — ideal for hackathon demos and production deployments alike.



Demo Mode

- Pre-generated mock findings
- Simulated SSE trace streams
- No API keys required
- Instant setup for presentations



Live Mode

- Azure OpenAI GPT-4o integration
- Real-time security analysis
- Authentic vulnerability findings
- Production-ready deployment



Automatic Mode Switching: The system detects API key availability and seamlessly switches between Demo and Live modes — no manual configuration needed.

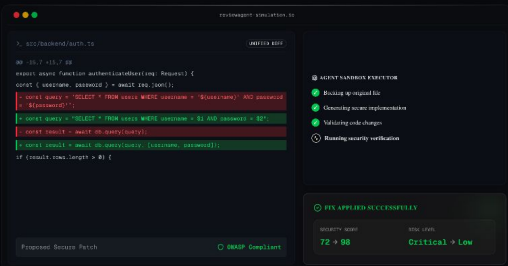
V2.4 ENTERPRISE SECURITY AUDIT

Automate Pull Request Reviews At AI Security Speed

An autonomous AI agent that audits code changes, simulates security checks against OWASP guidelines, and applies sanctioned remediations before you merge.

Launch Live Dashboard

Watch Interactive Demo



CORE CAPABILITIES

Enterprise Security Intelligence

Everything you need to secure your production build automatically inside your Pull Request lifecycle.



Vulnerability Scans

Scan codebases for CVE, exploits, secrets, misconfig, and infrastructure loopholes based on OWASP Top 10 guidance.



One-Click Remediation

Auto AI open remediation guidelines that backup files, write secure patches, test code logic, and apply fixes automatically.



Telemetry Dashboard

Comprehensive dashboard for vulnerability metrics, remediation success rates, system performance, and risk scoring analysis.



Webhook Integrations

Seamless integration with your CI/CD pipeline, issue trackers, and project management tools for automated security checks.

450k+

LINES AUDITED

99.4%

AUDIT REVIEW ACCURACY

8.2s

AVERAGE TURNAROUND

1,200h

MONTHLY SAVED

COMPATIBLE ECOSYSTEM

Engineered for Modern Deployments

Landing Page Experience

The entry point to ReviewAgent AI — designed to communicate value and onboard teams in seconds.

Product Introduction

Clear, concise messaging on AI-powered security review capabilities.

Console Simulator

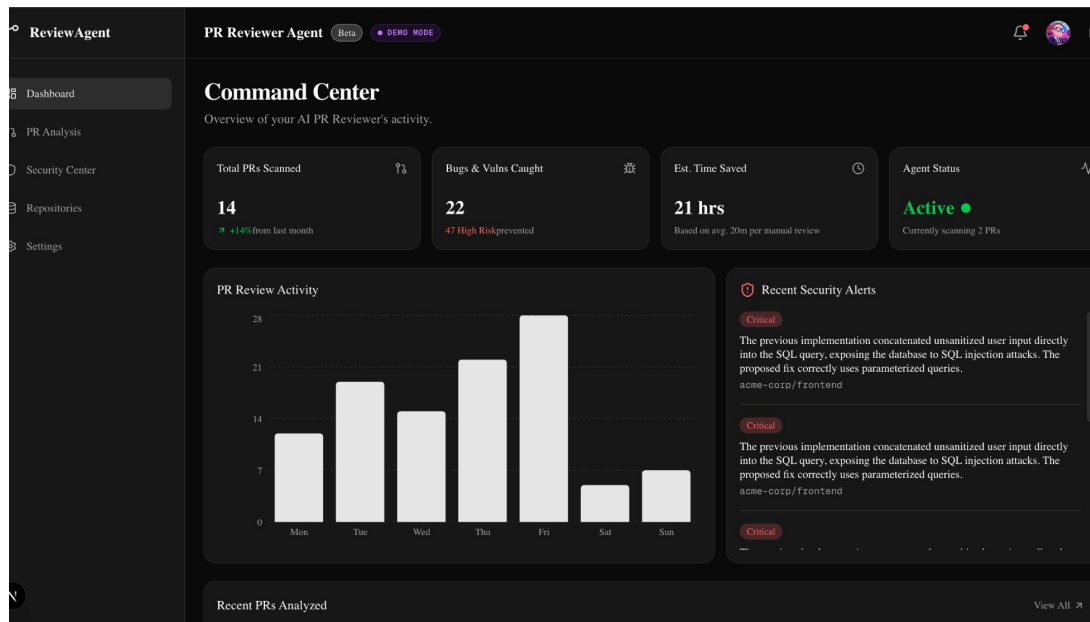
Interactive demo experience with live mode indicator.

Quick Dashboard Access

One-click entry to the security intelligence dashboard.

Security Intelligence Dashboard

A centralized command center for repository activity, security findings, and pull request review status — giving engineering and security teams a real-time view of their security posture.



→ Repository Overview

Unified view of all connected repositories and their status.

→ Security Metrics

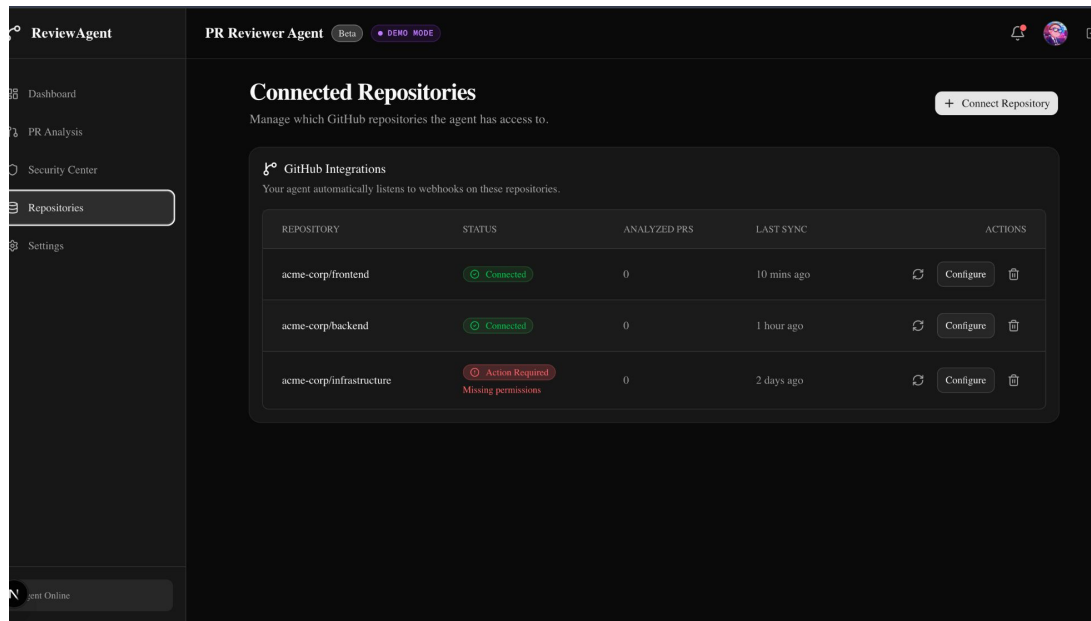
Live risk statistics and vulnerability trend charts.

→ AI Review Status

Real-time tracking of ongoing and completed PR reviews.

Repository Management

Connect, monitor, and manage source code repositories for fully automated security review workflows.



→ Repository Registration

Seamlessly connect Git repositories with OAuth-based authorization.

→ Status Monitoring

Live sync status and health indicators for every repository.

→ PR Tracking

Automated pull request detection and review queue management.

Security Center

Full visibility into vulnerabilities, risk categories, and security trends across all repositories — enabling teams to prioritize what matters most.



Critical

Immediate action required

High

Address within sprint

Medium

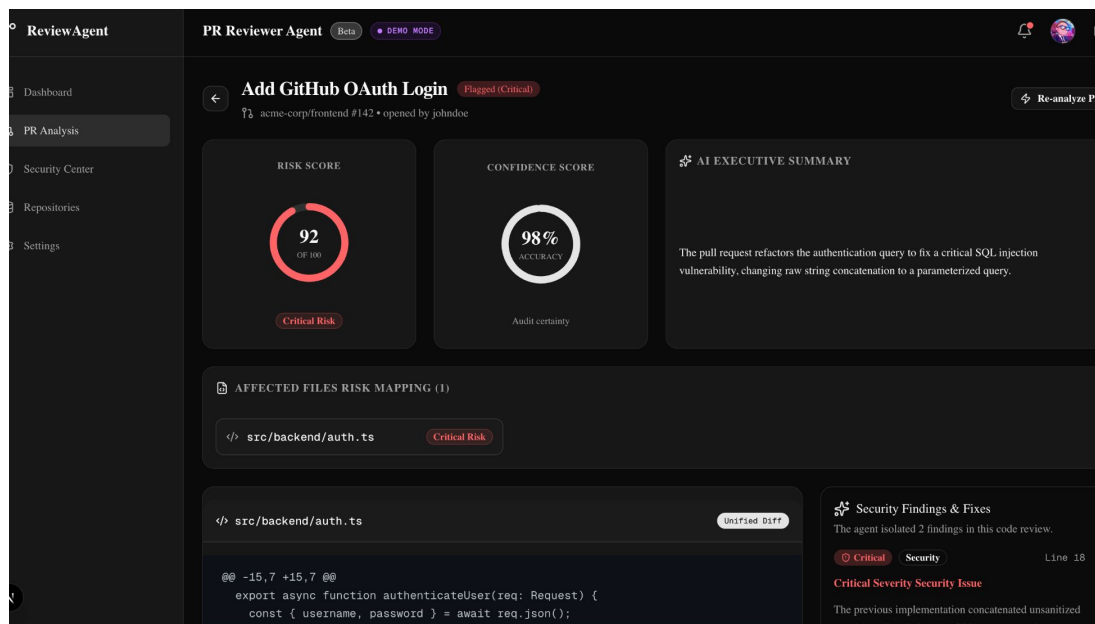
Schedule for review

Low

Track and monitor

AI-Powered Pull Request Analysis

The core engine of ReviewAgent AI — combining code diff analysis with explainable AI reasoning to deliver precise, actionable security findings.



Code Diff Viewer

Inline security annotations on every code change.



AI Reasoning Trace

Transparent explanation of every finding and risk score.



Suggested Code Fixes

Auto-generated remediation guidance developers can apply immediately.

Analysis Output: Explainable & Actionable

Every review delivers a structured, confidence-scored report — not just a flag, but a full reasoning trace and fix recommendation.

92%

Risk Score

Severity: Critical

97%

Confidence Score

AI certainty on finding

3

Fix Suggestions

Auto-generated remediation

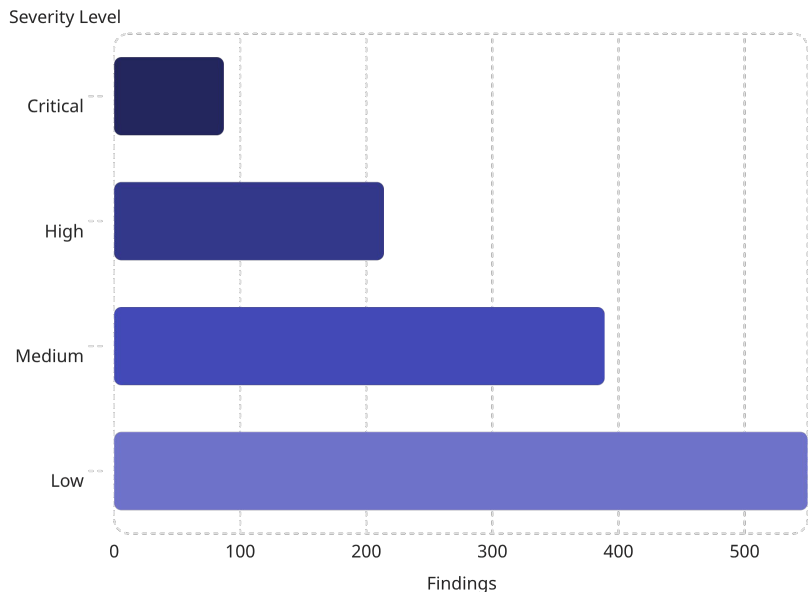
<1s

Review Time

Per pull request

 All findings include a full AI reasoning trace — security teams understand **why** a vulnerability was flagged, not just that it exists.

Security Findings Breakdown



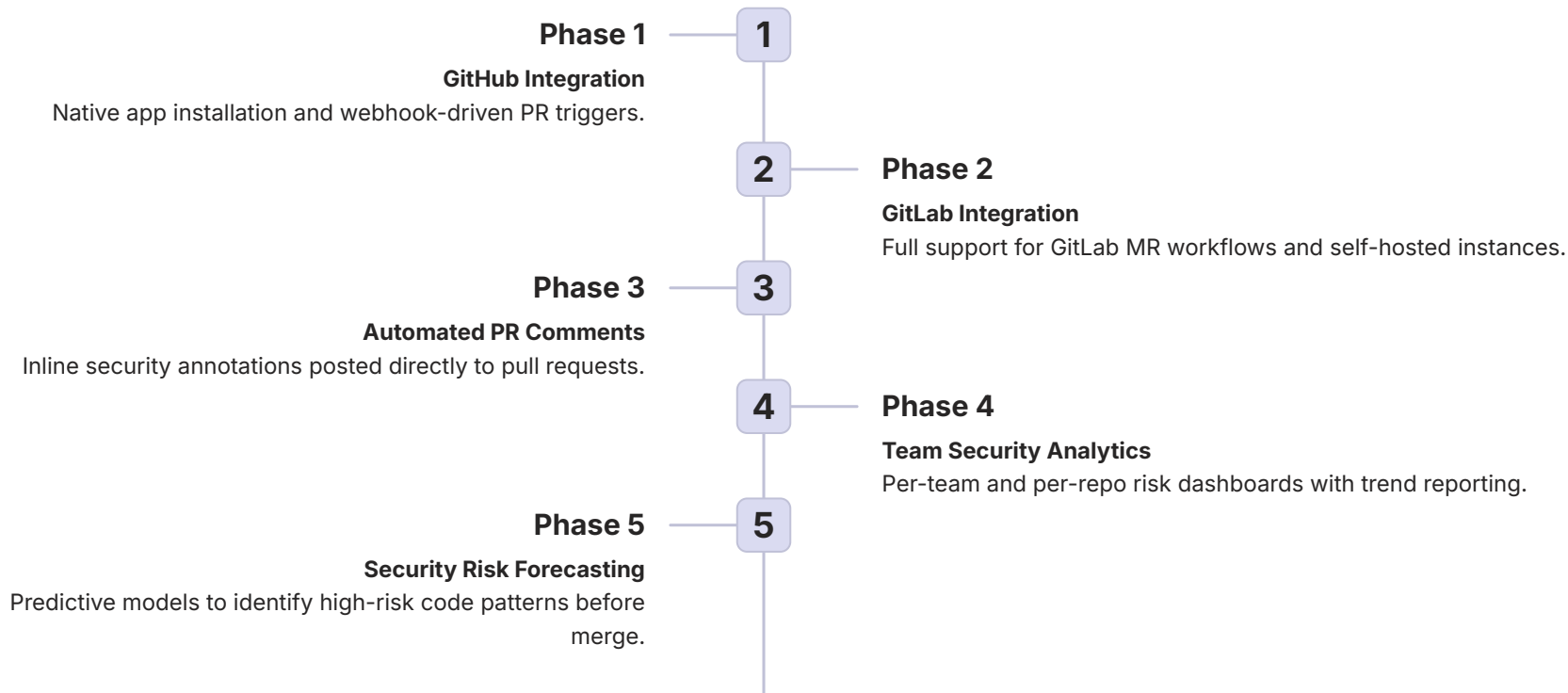
What the Data Tells Us

ReviewAgent AI surfaces findings across the full severity spectrum, enabling teams to triage efficiently and focus engineering effort where risk is highest.

- **Critical & High** findings are surfaced immediately in the Security Center
- **Medium & Low** findings are tracked for scheduled remediation
- Trend analytics show whether security posture is improving over time

Future Roadmap

A phased rollout expanding platform coverage, automation depth, and predictive security intelligence.



Thank You!

